

LICENSING OFFICER

 Review queue

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3 need review · 0 waiting on operators · 4 decided

Refreshed just now · updates every 30 s



PermitFlow

Modernising regulatory licensing workflows

A working product for the Food Establishment Licence: submit, review, feedback, resubmit and decide, with an advisory document check.

Needs review 3

Waiting on operator 0

Decided 4

All 7

REFERENCE

BUSINESS · APPLICANT

STATUS · CHECKS

PF-2026-001009

Revision 1 · 2 open

Kopi & Kaya Toast House Pte. Ltd.

Tan Wei Ling · 10 Jalan Besar #01-12 · just now

● Under Review

just now

● 2 to check

Submitted 19 Sep 2026

Continue review →

PF-2026-001001

Revision 1

Scenario Kopi House Pte. Ltd.

Tan Wei Ling · 10 Jalan Besar #01-12 · 14 h ago

● Under Review

14 h ago

● 4 to check

Submitted 19 Sep 2026

Continue review →

PF-2026-001000

Revision 1

Scenario Kopi House Pte. Ltd.

Tan Wei Ling · 10 Jalan Besar #01-12 · 19 h ago

● Under Review

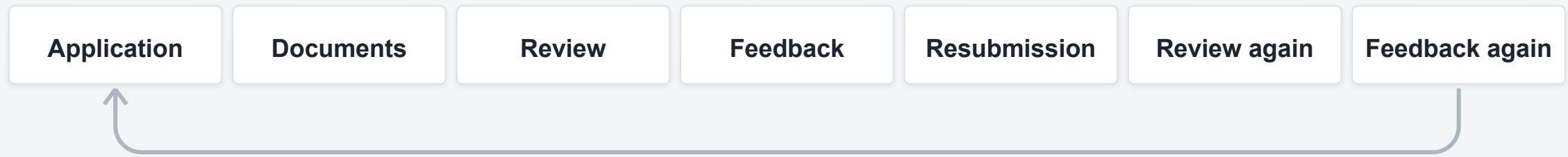
19 h ago

● 4 to check

Submitted 18 Sep 2026

Continue review →

Licence applications go round in a slow, lossy loop, and each turn costs both sides time and trust.



- 01 Incomplete or incorrect submissions cause repeated rejection cycles.
- 03 The whole application is re-entered on resubmission and data is lost between rounds.
- 05 The same comments are rewritten for the same common problems.

- 02 Feedback is not tied to a section or a document, so the business does not know what to fix.
- 04 The officer cannot see what changed, or whether an earlier issue was actually resolved.
- 06 Applications get lost between stages and the record of what happened is incomplete.

Before: a pile, a letter with no reason, the whole thing again, and an officer comparing by hand.



1

The pile

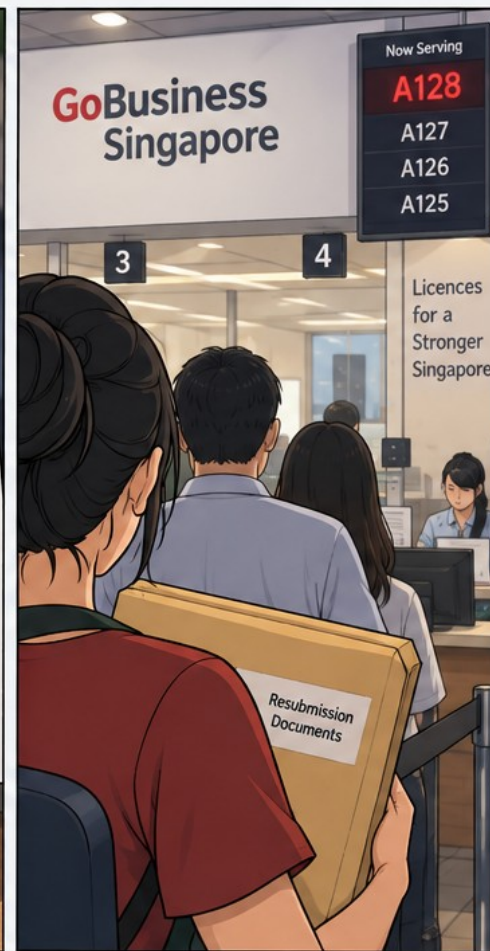
Incomplete documents and months of waiting.



2

The letter

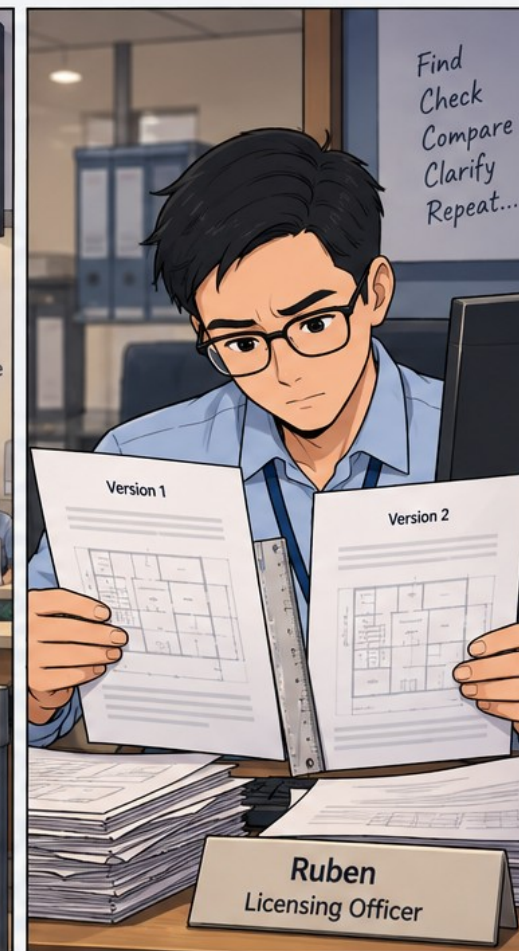
The application is returned as incomplete.



3

Everything again

Queue up and submit everything again.

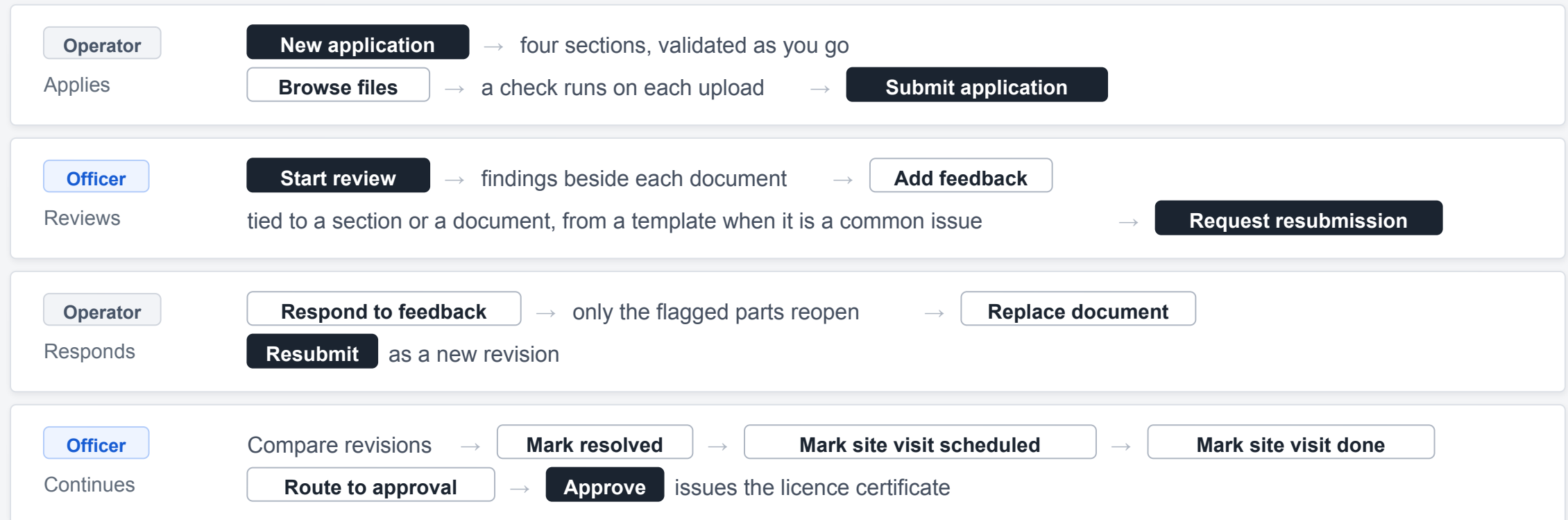


4

The officer's hunt

Comparing documents to find what's missing.

One continuous workflow: submit, review, feedback tied to a section or a document, then resubmit only what was flagged.

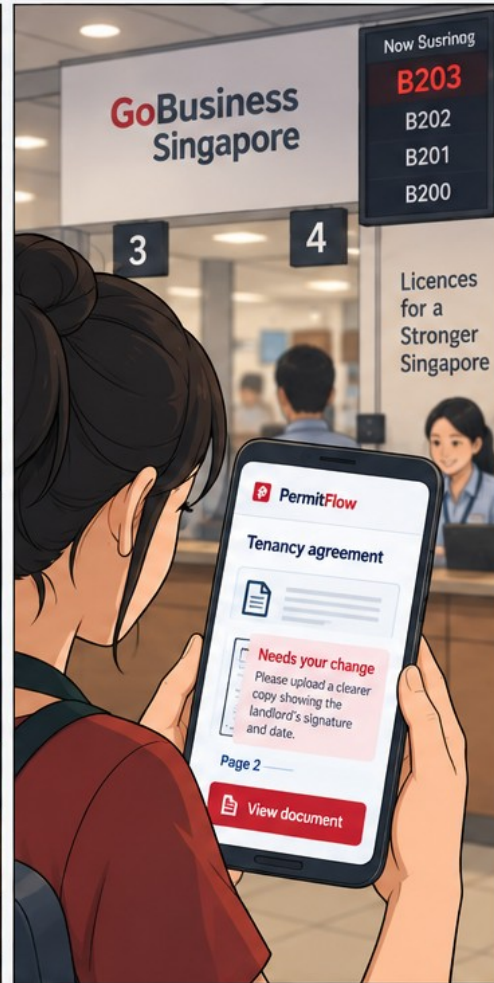


Every dark or outlined chip is a control label in the product, spelled as it appears.

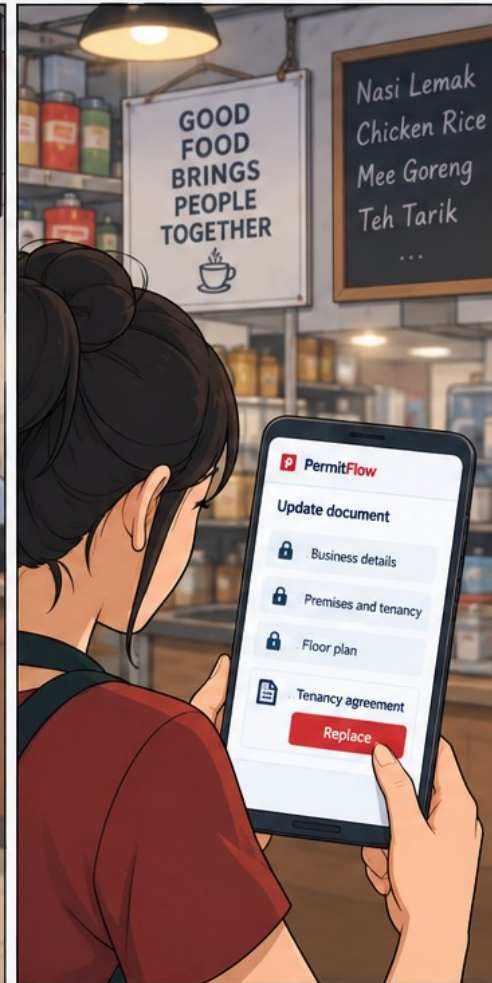
After: the same four moments with PermitFlow, and each one is shorter.



1 A clearer path
See your progress at a glance.



2 Clear, specific feedback
See exactly what to fix.



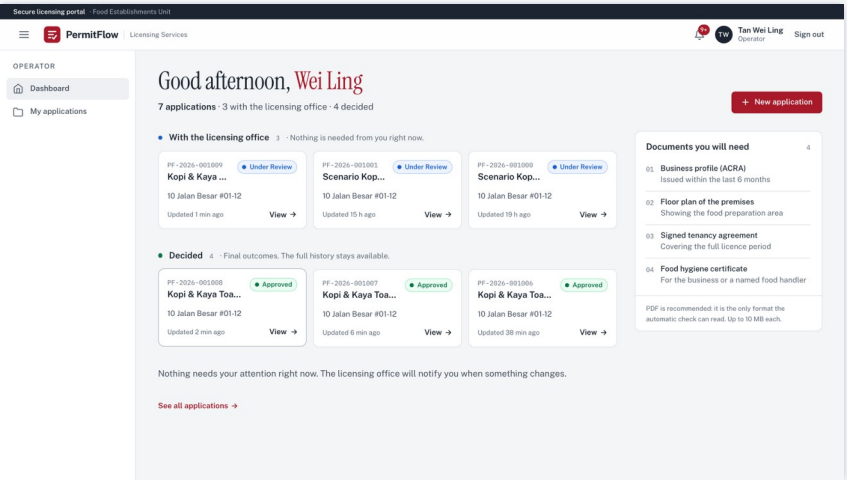
3 Only what's needed
Update just the flagged part.



4 Ready for approval
Issues resolved, licence within reach.

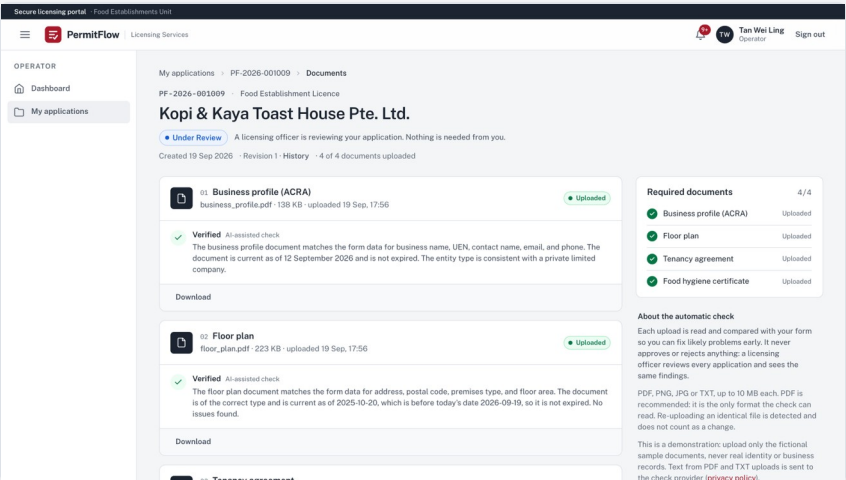
The product runs one workflow for two personas and keeps every step traceable.

06 / 13



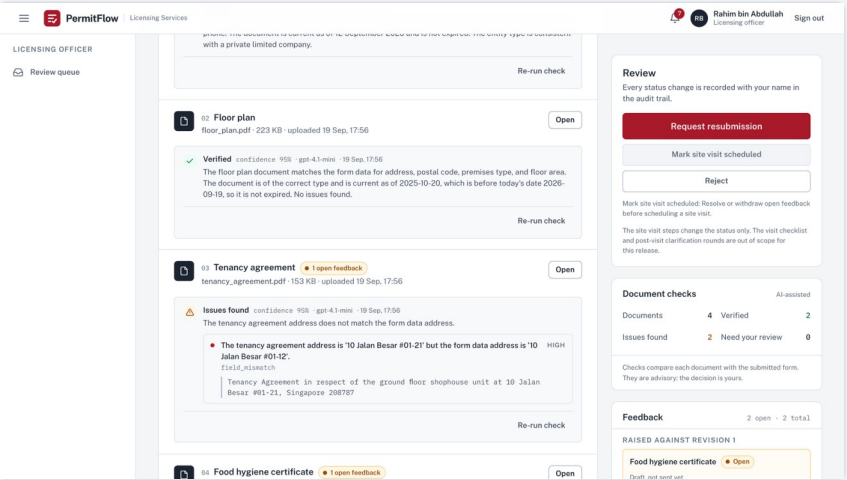
Operator

Dashboard: applications grouped by who needs to act



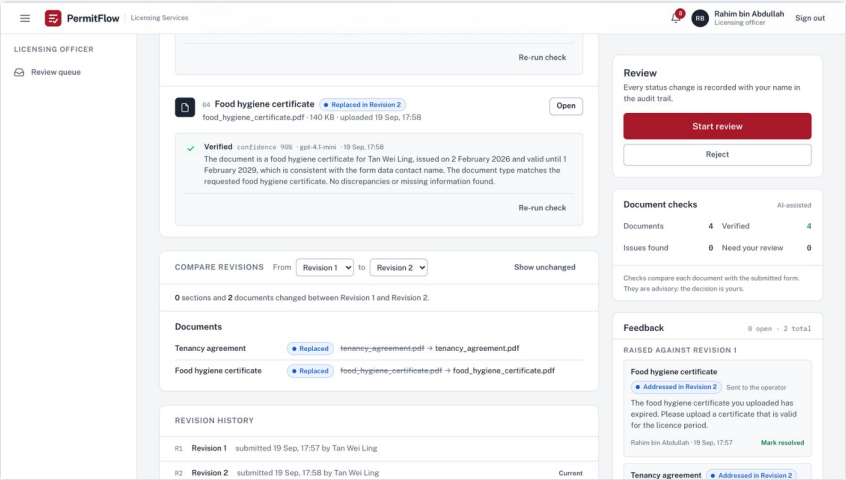
Operator

Documents: a check result on each upload, in plain language



Officer

Case: findings with evidence beside each document, feedback rail on the right



Officer

Compare revisions: what changed, and each item's resolution

The AI assists. Humans decide.

Per upload, before submission

Advisory check on every document

- Extract the text; unreadable files are reported without a model call
- Compare the document with the form: name, UEN, address, dates
- Flag issues with a closed issue code
- Quote the evidence from the document
- Self-reported confidence; below 0.6 becomes Needs your review

What the officer sees

Officer

- Outcome: Verified, Issues found, Needs your review
- Each issue with its code, severity and evidence quote
- Confidence and model on the card; prompt version in the audit event and the trace

Re-run check at any time

What the officer decides

Every decision is a person's, and audited

- The findings never change a status or write feedback
- The Approve dialog warns about unresolved findings but never blocks
- The state machine has no AI actor

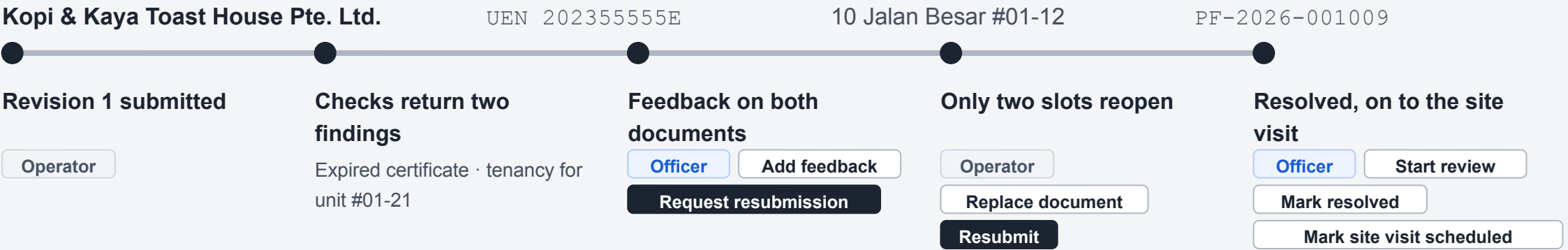
Deterministic by design, not AI

Never touched by the model

- Feedback templates
- Revision compare
- Status changes
- The prompt-injection rule, applied before the verdict is read

Operator sees the outcome, a plain explanation and each issue in plain words. **Never the evidence quote, never the confidence.**

Two planted problems, one feedback round, and the record shows exactly what changed.



DOCUMENTS AND CHECK RESULTS

Business profile (ACRA)

business_profile.pdf

● Verified

Floor plan

floor_plan.pdf

● Verified

Tenancy agreement

Replaced in Revision 2 · tenancy_agreement.pdf

● Verified

Food hygiene certificate

Replaced in Revision 2 · food_hygiene_certificate.pdf

● Verified

FEEDBACK RAISED AGAINST REVISION 1

Food hygiene certificate

Template: Food hygiene certificate has expired

● Resolved

Tenancy agreement

The tenancy agreement names unit #01-21, but the application and the business profile give #01-12

● Resolved

Compare revisions

From Revision 1 to Revision 2

0 sections and 2 documents changed

Status

● Site Visit Scheduled

Six principles, each enforced by code.

01 **Continuous workflow**

One application, any number of rounds. Fourteen states with actors, guards and labels as a table; every status change goes through one function.

02 **Immutable revisions**

Every submission is a snapshot. The compare view is a read-time diff, computed and never stored, so nothing is lost between rounds by construction.

03 **Contextual feedback**

Every item is tied to a section or a document. "Only the flagged parts" is an authorization rule: the server answers 403, the UI walks the operator through exactly those targets.

04 **AI-assisted, never AI-decided**

The verifier writes only to `verification_runs`. The state machine has no AI actor. Injection is handled by a deterministic rule before the verdict is read.

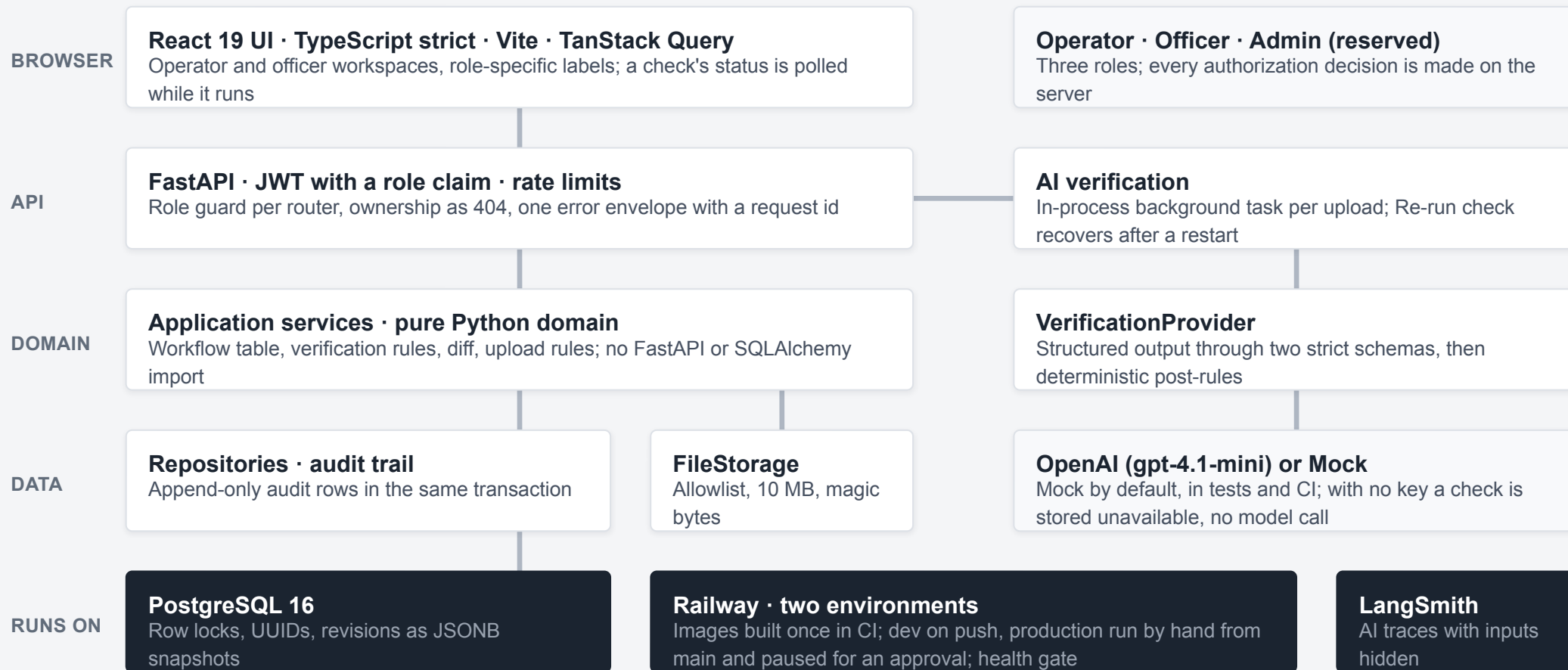
05 **Same-transaction audit**

A status change, its audit row, the operator's notification and, on approval, the licence certificate either all happen or none do.

06 **Role-aware by construction**

Authorization is server-side on every route with an authorization test per endpoint. Operators receive their own labels and never an internal status code.

A modular monolith with an explicit state machine, an advisory AI module and an audit trail written in the same transaction.



Checked by tests, gates and a written record.

748 backend tests **152** frontend tests **8** Playwright specs **96 % and 80.6 %** coverage **588** state-machine combinations **22** threats **70** stories, 4 deferred

- ✓ **Tests.** Backend on a real PostgreSQL; Playwright journey and scenarios in CI against the full stack.
- ✓ **State machine.** 588 transition combinations over 14 states and 3 actors; every invalid one rejected.
- ✓ **Authorization.** Server-side on every route; one test per endpoint; ownership answered as 404.
- ✓ **Validation.** Strict schemas at the HTTP boundary and on the model's output; one error envelope with a request id.
- ✓ **Audit.** Append-only rows in the same transaction as the change; actor and time on every event.
- ✓ **CI gates.** Lint, types, coverage thresholds, the six-stage AI gate, accessibility, secret and dependency scans.
- ✓ **AI evaluation.** 14 golden cases and a 21-run fairness check on the live model; nightly and on change.
- ✓ **Delivery.** Images built once; production behind an approval; health gates after every deploy; request ids everywhere.

Known gaps first, then what production would need.

01 **Use case 3: the site-visit checklist**

Statuses and transitions exist and are tested; the checklist, capture screen and clarification round are not built.

03 **Object storage with a virus scan**

S3-compatible storage with signed URLs behind the existing storage interface.

05 **Identity**

Registration, password reset, MFA and SSO; seeded accounts today.

07 **AI evaluation and assurance**

A labelled set from officer overrides, broader red-teaming, calibration, multilingual injection defence.

09 **Delivery and backups**

Scheduled backups with a tested restore; a staging rehearsal for migrations; image scanning.

02 **A queue and a worker for the AI checks**

So checks survive deploys and scale on their own; today they run in-process.

04 **Frontend and edge hardening**

httpOnly cookie sessions, CSP nonces, rate-limit windows shared across instances.

06 **Data protection**

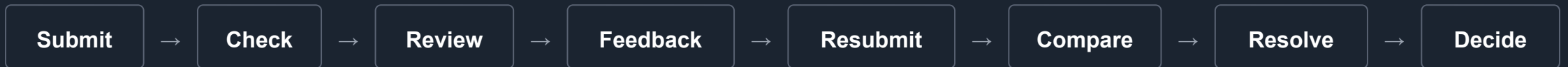
Retention and redaction for document text sent to the model; an in-region provider.

08 **The admin epic**

Oversight dashboard, AI health, audit feed, user management; role and routes reserved.



From submission to resolution: one traceable licensing workflow.



A working MVP built with production practices, in a three-day engineering constraint.

permitflow.space
github.com/SuhaasNv/permitflow